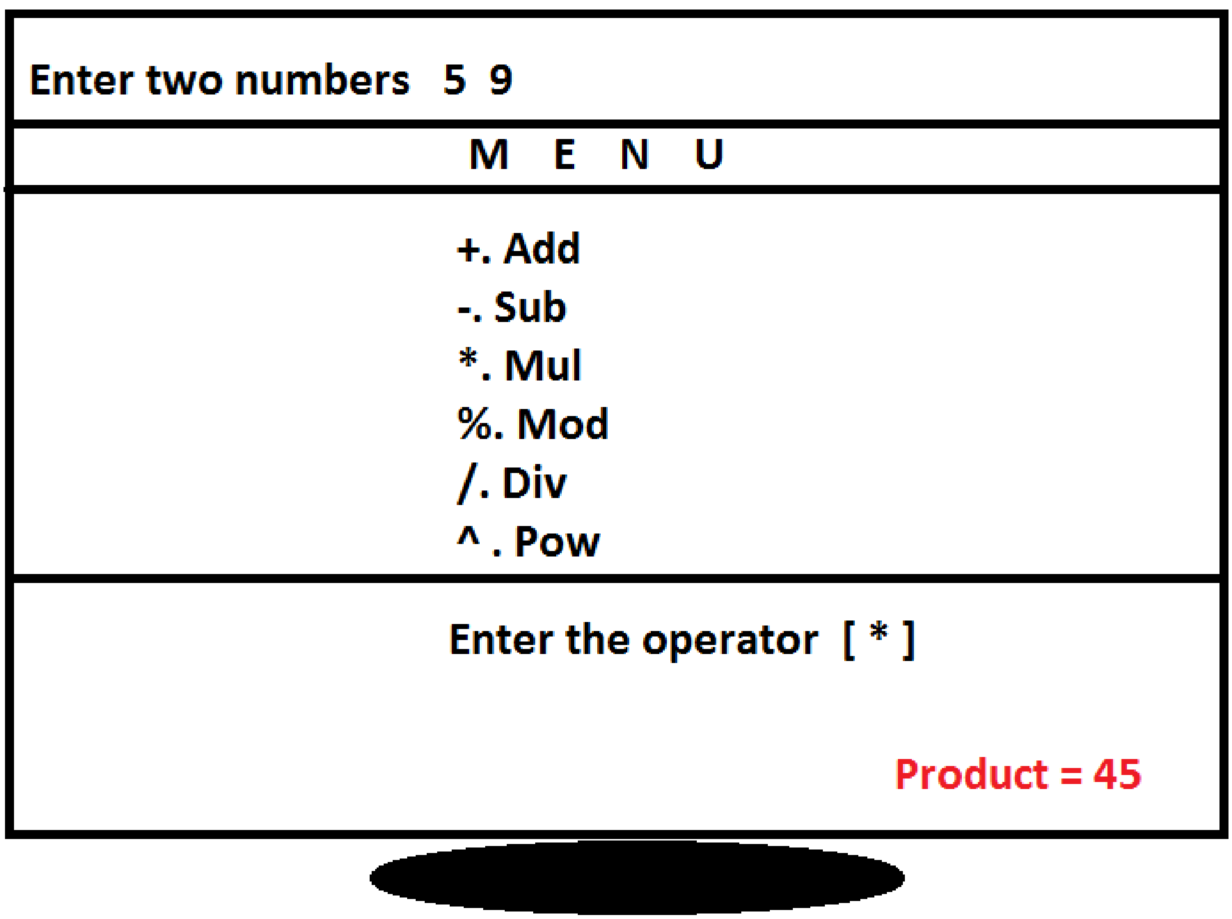
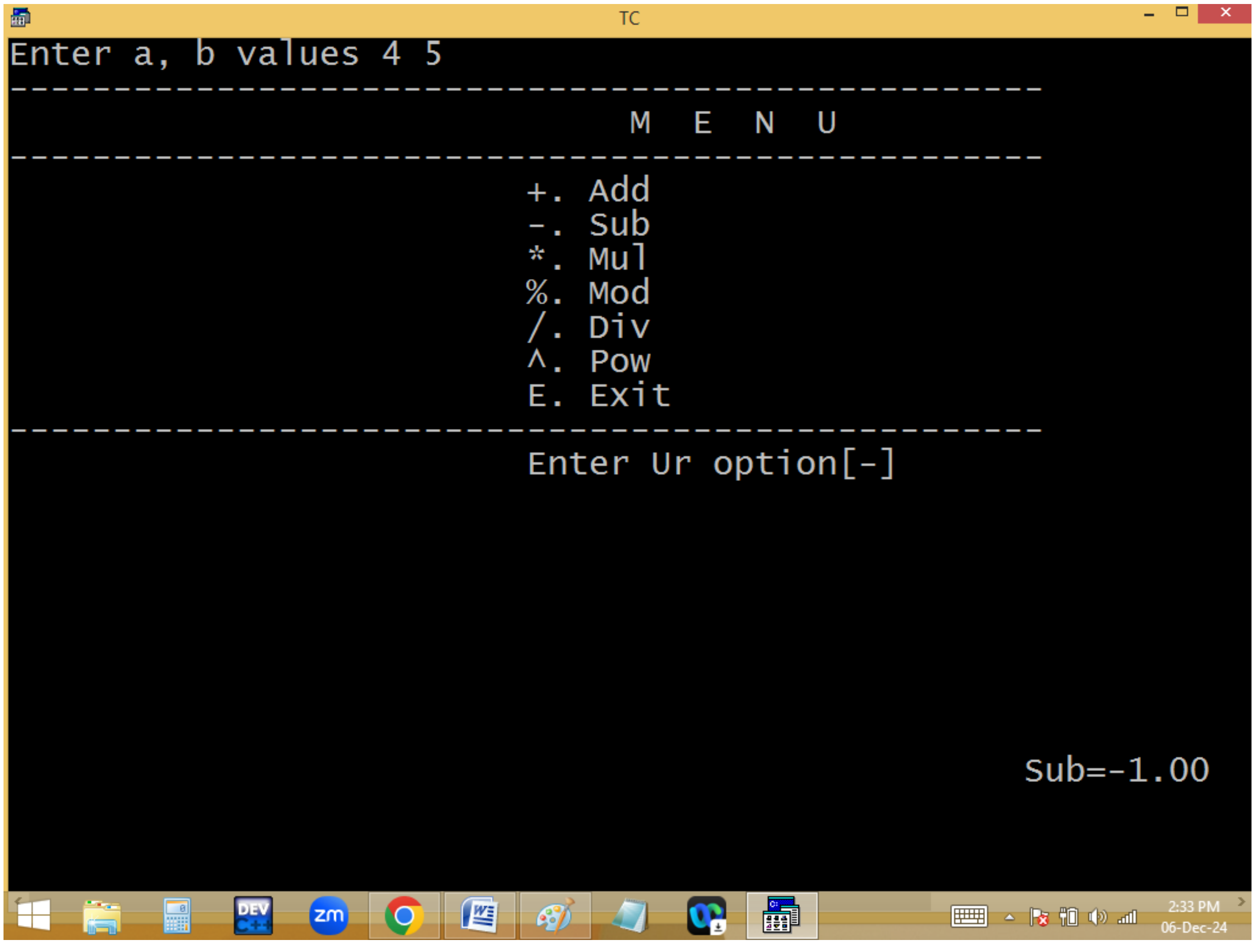
Read two numbers and perform arithmetic operation using menu with switch:

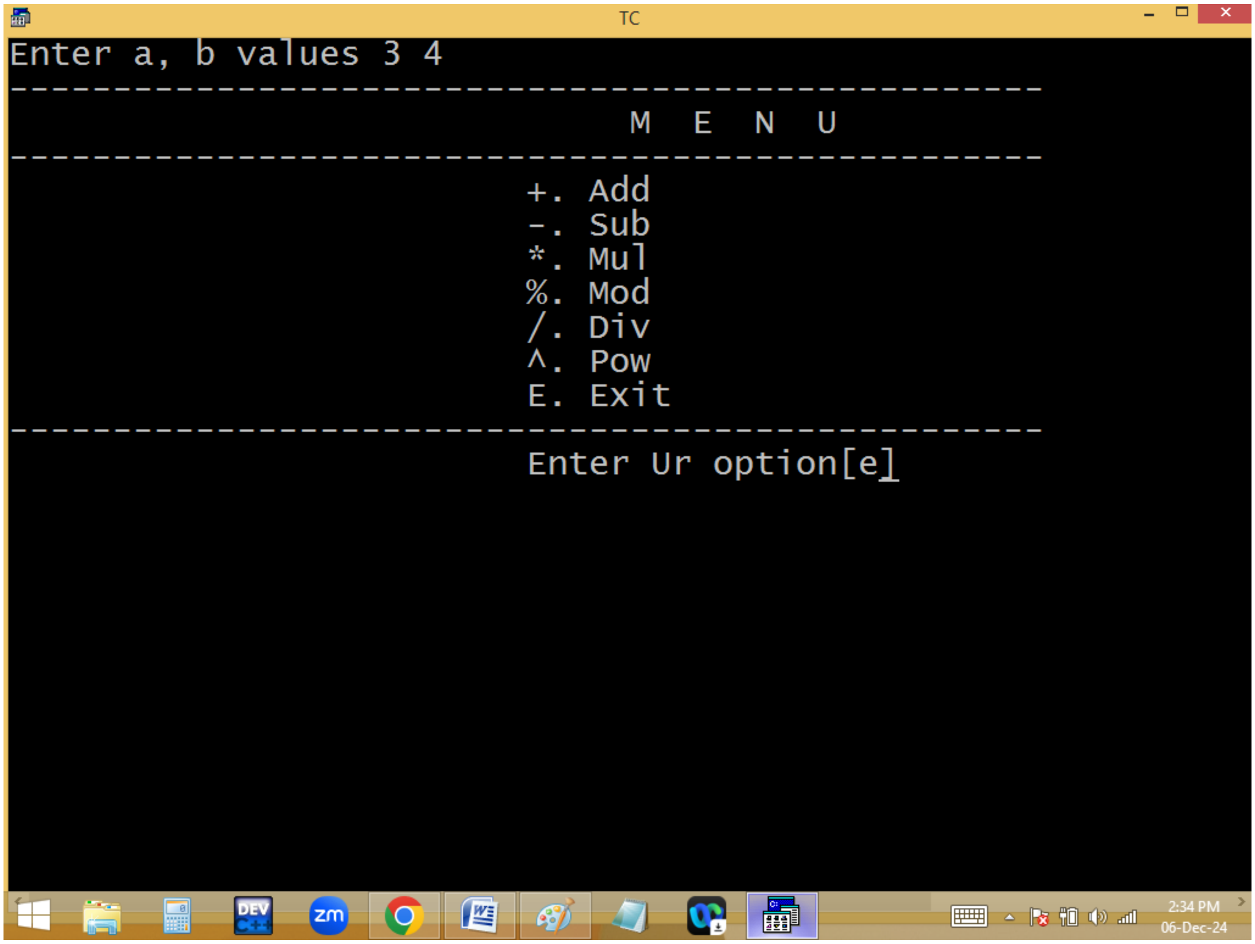
```
#include<stdio.h>

#include<conio.h>

#include<math.h>

void main()

{

float a, b; char op;

abc:

clrscr();

printf("Enter a, b values "); scanf("%f %f",&a, &b);

puts("----------------------------------------------");

puts("\t\t\t     M  E  N  U");

puts("----------------------------------------------");

puts("\t\t\t +. Add");

puts("\t\t\t -. Sub");

puts("\t\t\t *. Mul");
```

```
puts("\t\t\t %. Mod");

puts("\t\t\t /. Div");

puts("\t\t\t ^. Pow");

puts("\t\t\t E. Exit");

puts("--------------------------------------------------");

printf("\t\t\t Enter Ur option[ ]\b\b");

flushall();

scanf("%c",&op);

gotoxy(50, 22);

switch(op)

{

case '+': printf("Sum=%.2f",a+b);break;

case '-': printf("Sub=%.2f",a-b);break;

case '*': printf("Mul=%.2f",a*b);break;

case '%': printf("Mod=%.2f",fmod(a,b));break;

case '/': printf("Div=%.2f",a/b);break;

case '^': printf("Pow=%.2f",pow(a,b));break;
```

```
case 'e': case 'E': return;

default: printf("Invalid Option");

}

getch();

goto  abc;

}
```




TC
Enter a, b values 4 5
M E N U
+. Add
-. Sub
*. Mul
%. Mod
/. Div
^. Pow
E. Exit
Enter Ur option[-]
Sub=-1.00
2:33 PM
06-Dec-24



LOOPS / ITERATIONS / REPETITIVE STATEMETNS

Loops are used to repeat a block/group of statements continuously until the given condition becomes false.

Loops reduce program size and improves performance.

In loops beginning and ending points are same.

C-Language supports basically 2 types of loops.

1. **Entry/pre controlled loops.**
2. **Exit/post controlled loops.**

In entry control loops, condition is tested first and it is true then only statements block is executed.

Under entry control loops we are having

i. **While loop**
ii. **For loop**

In exit control loop, the statements are executed first and later condition is tested.

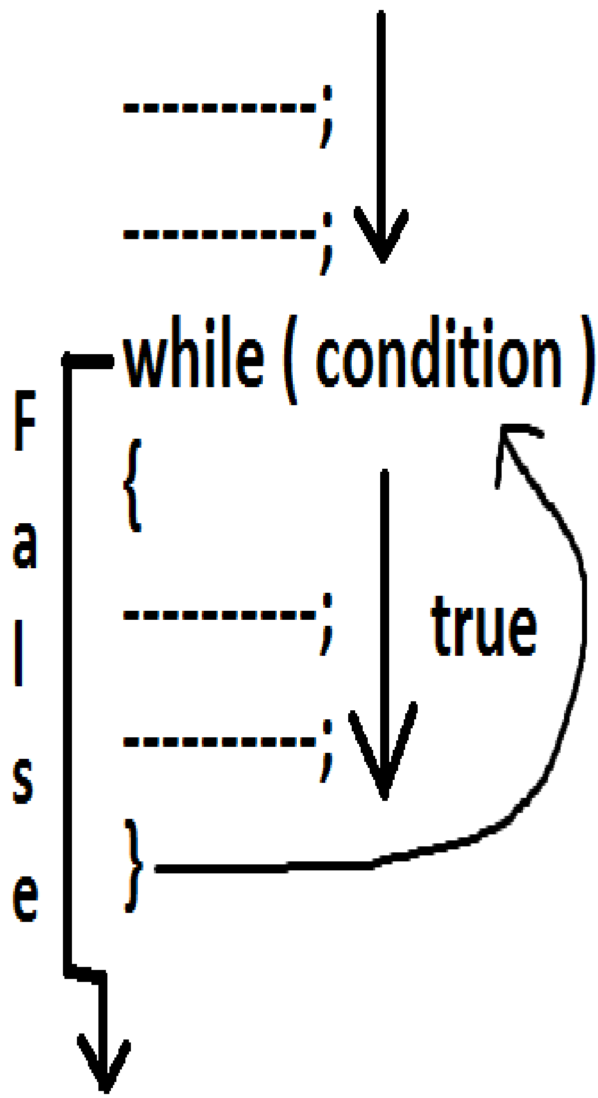
Under exit control loop we are having

i. **do while.**

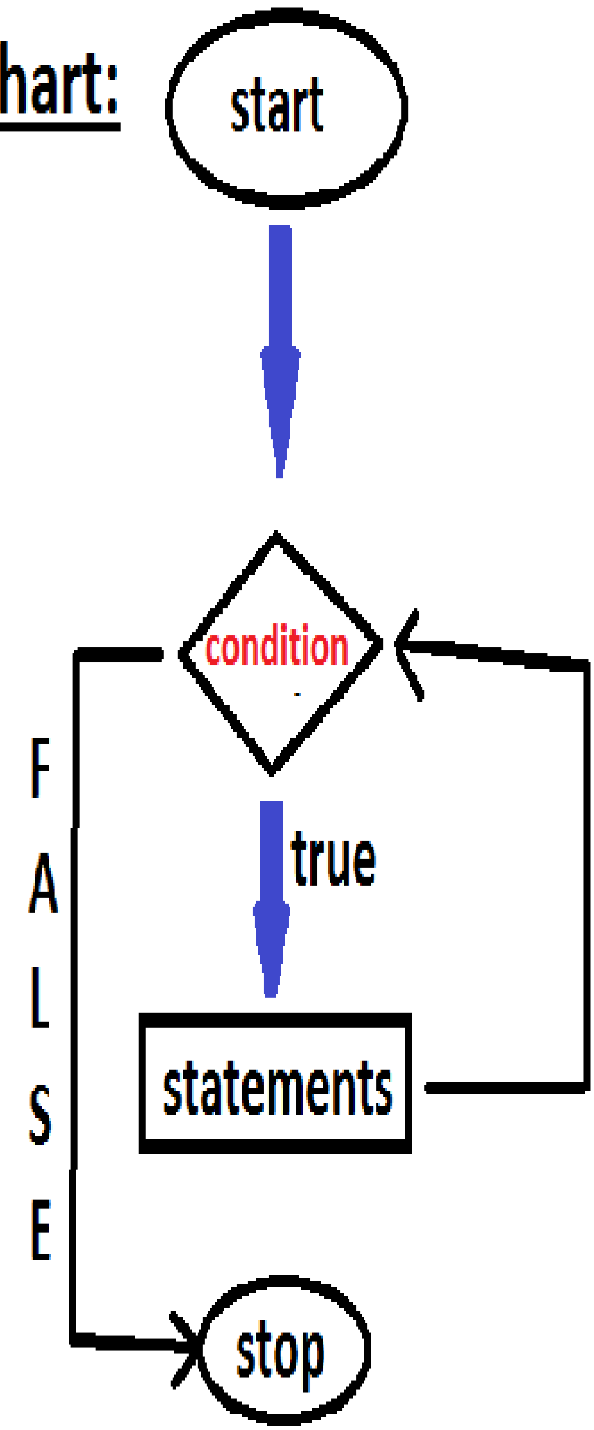
While loop:

- **while is a keyword.**
- **In while loop condition is tested first and it is true then only while block statements are executed. After executing while block statements, the program execution automatically shifted/jumped to while condition at the beginning. If it is true then once again the while block statements are repeated. Like this the process is continued until while condition becomes false.**
- **While is entry control loop.**

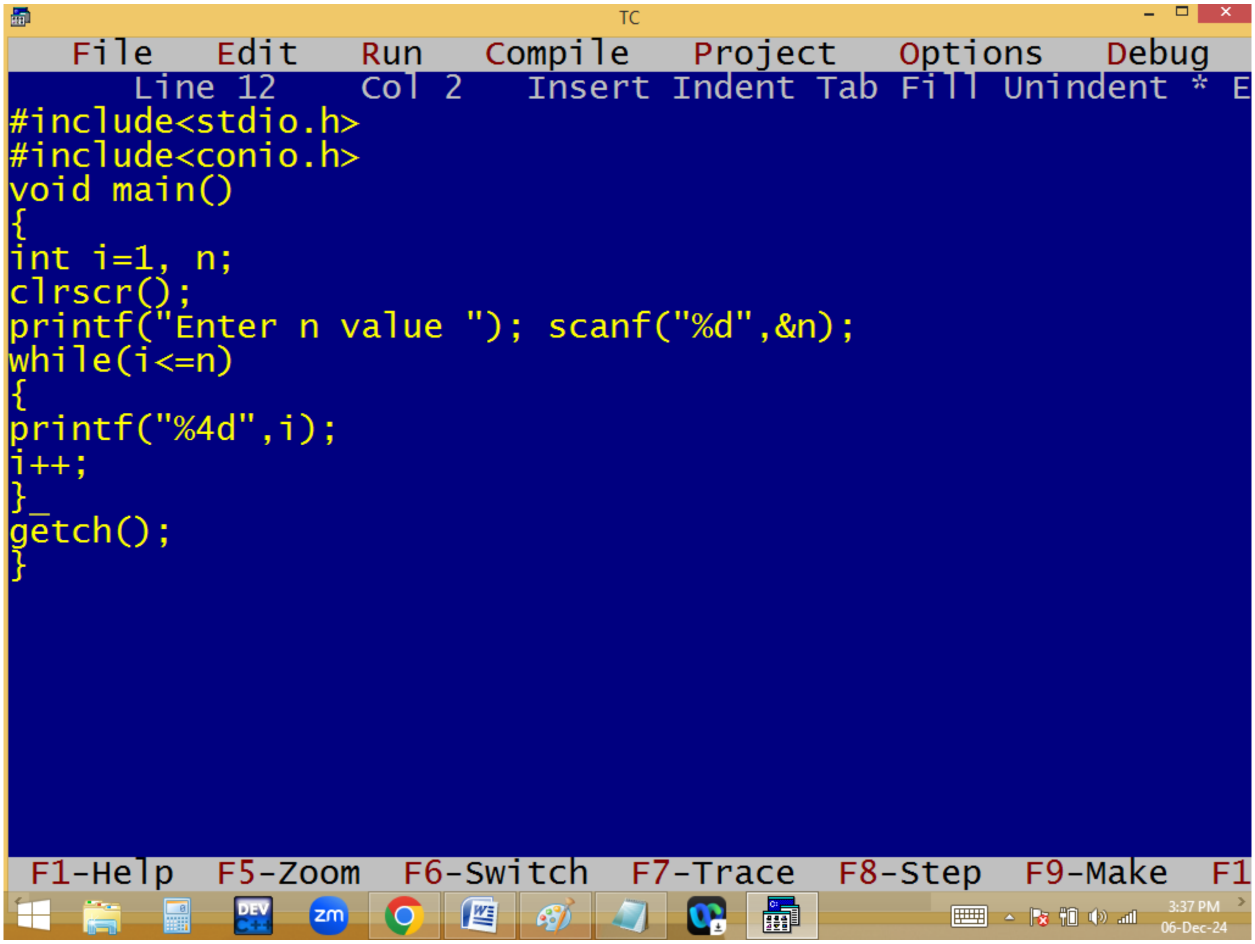
Syntax:

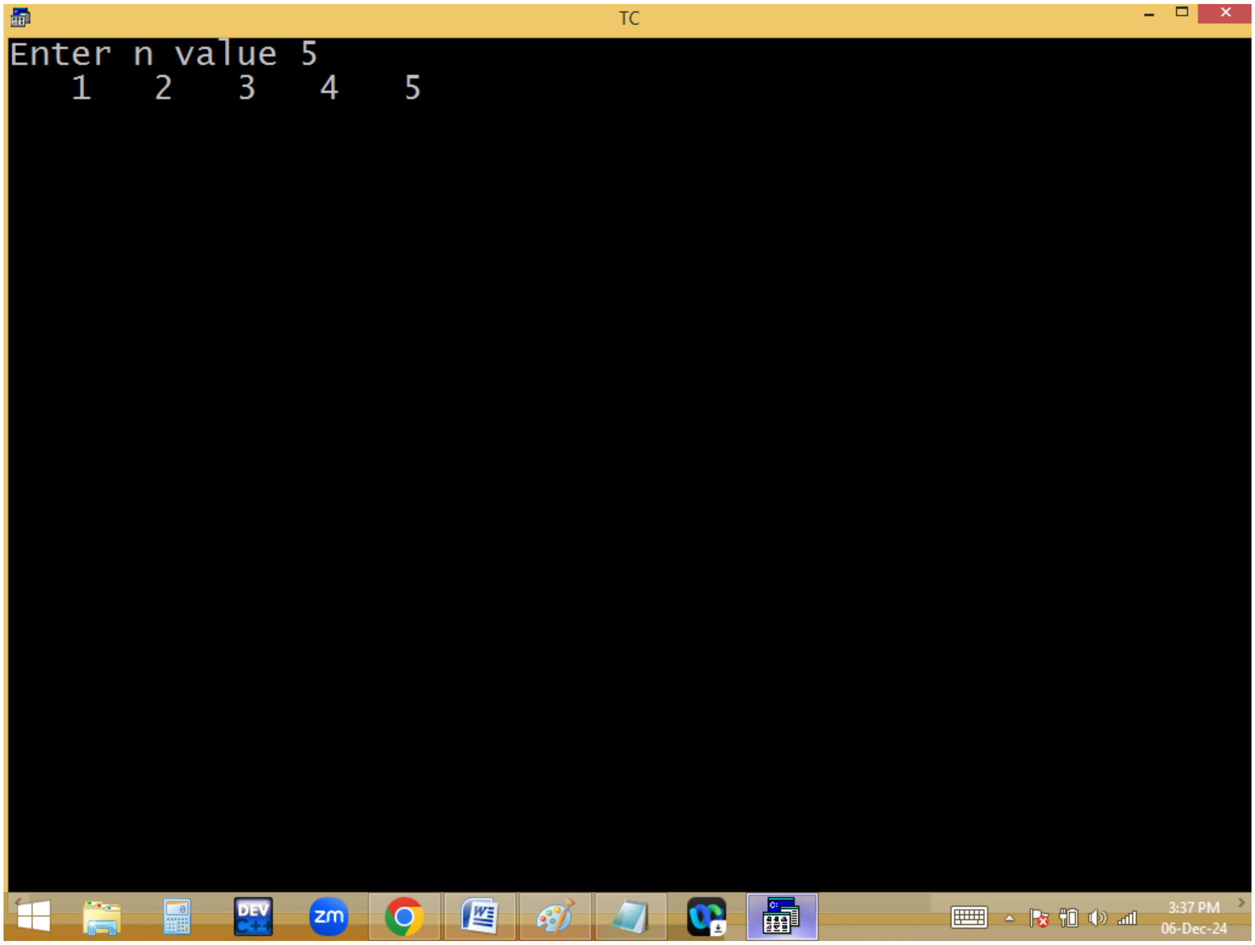


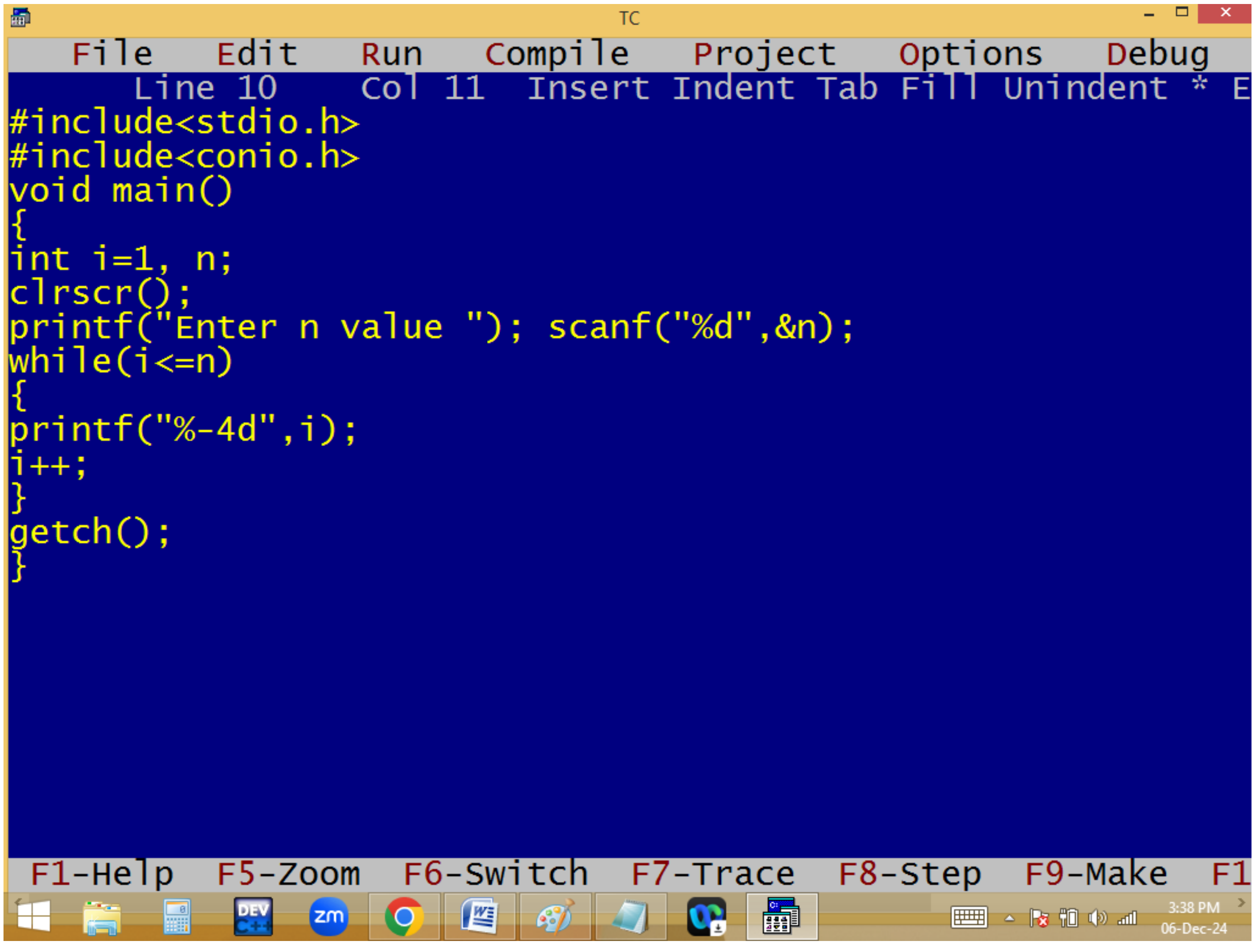
Flow chart:



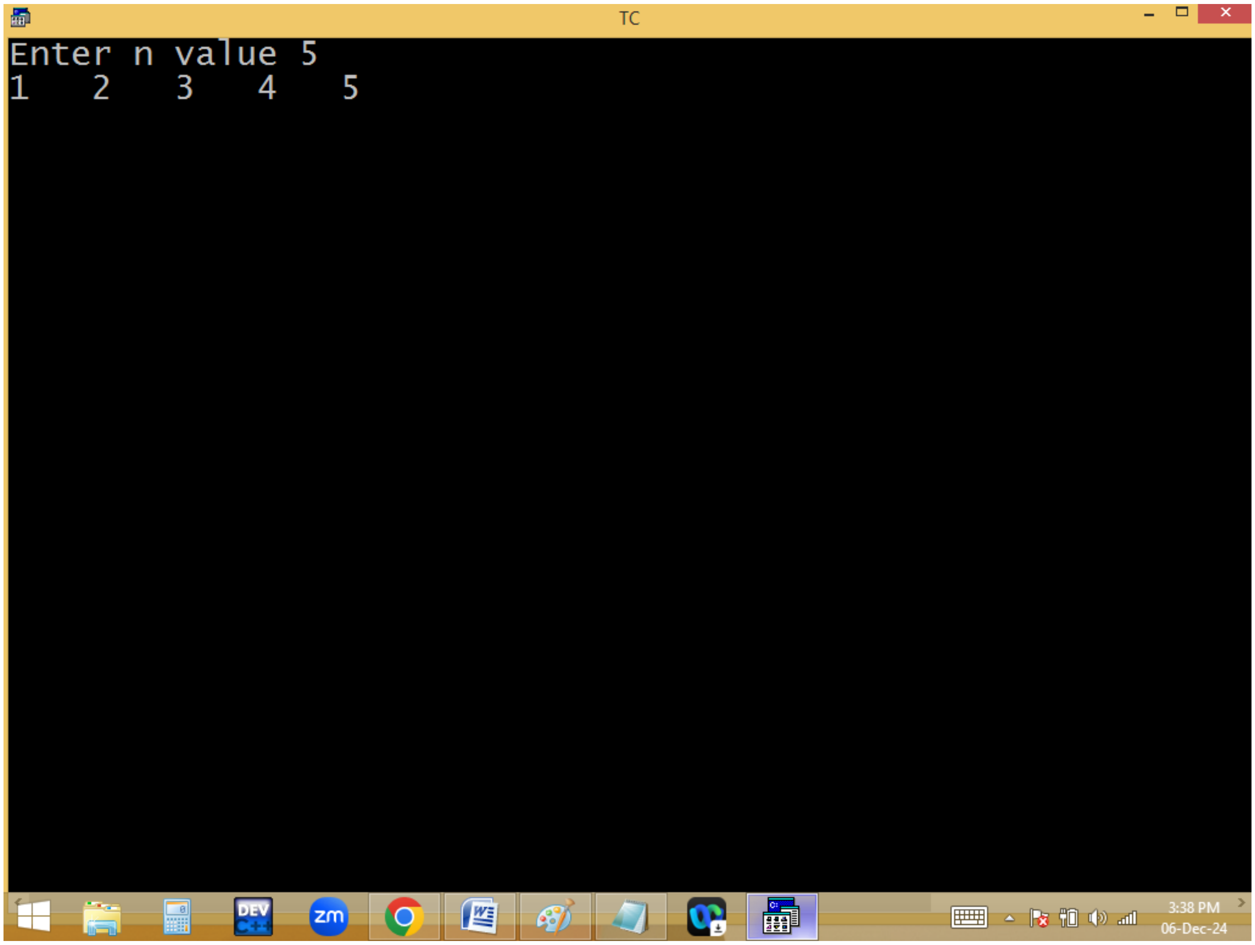
Eg: printing 1..n numbers

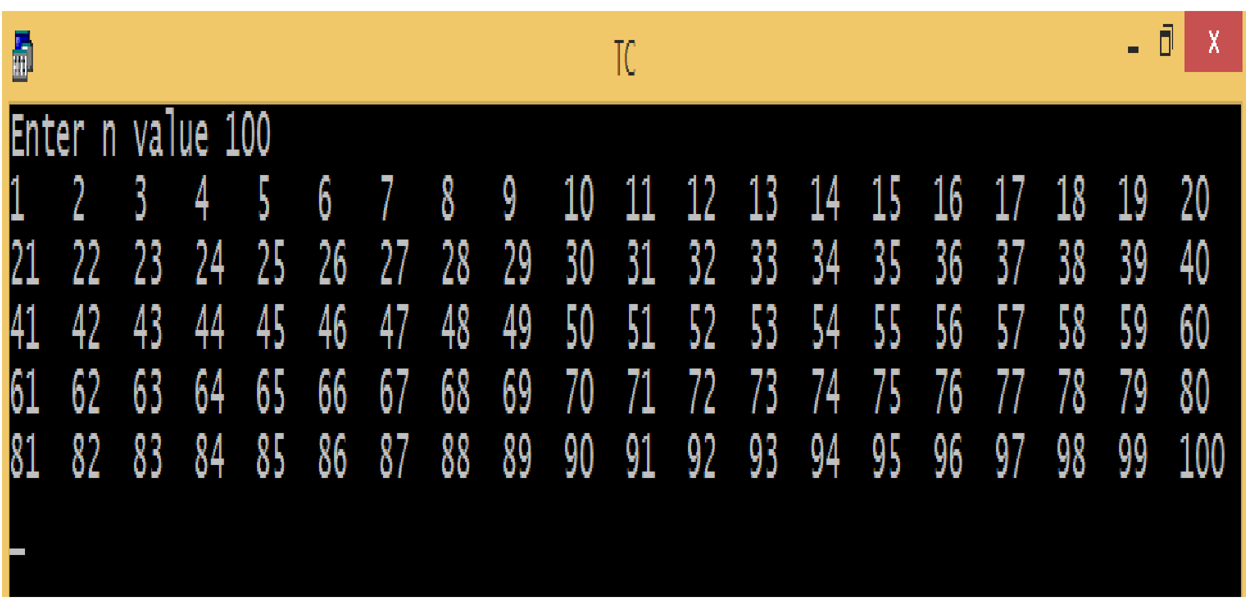



TC
Enter n value 5
1 2 3 4 5
3:37 PM
06-Dec-24



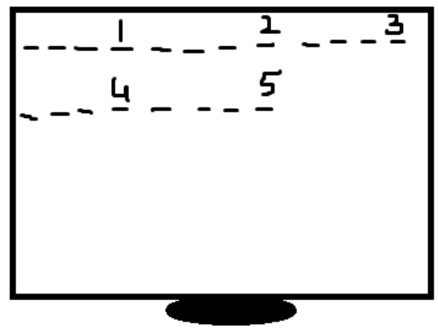

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)
{
printf("%-4d",i);
i++;
}
getch();
}
```

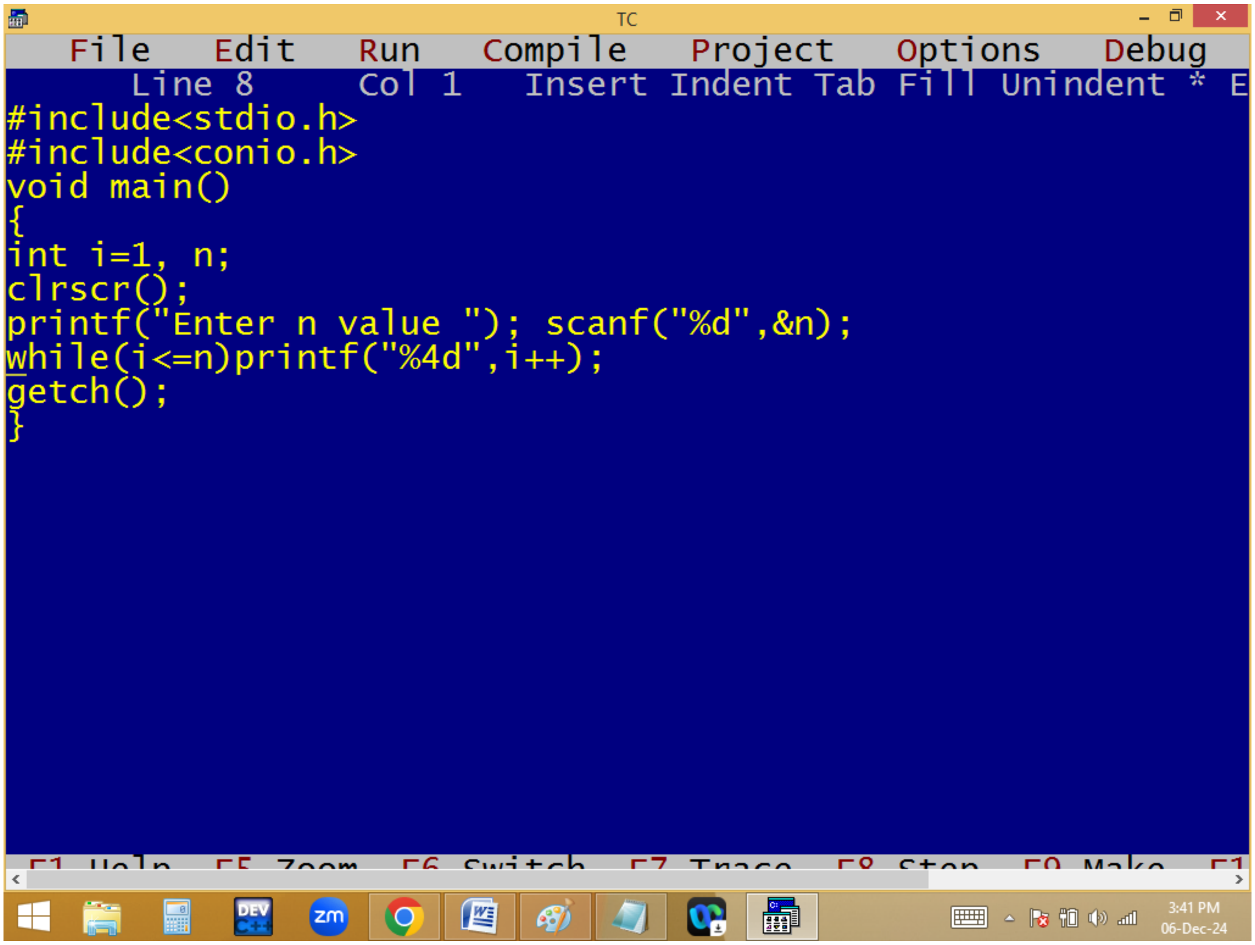

TC
Enter n value 5
1 2 3 4 5
3:38 PM
06-Dec-24

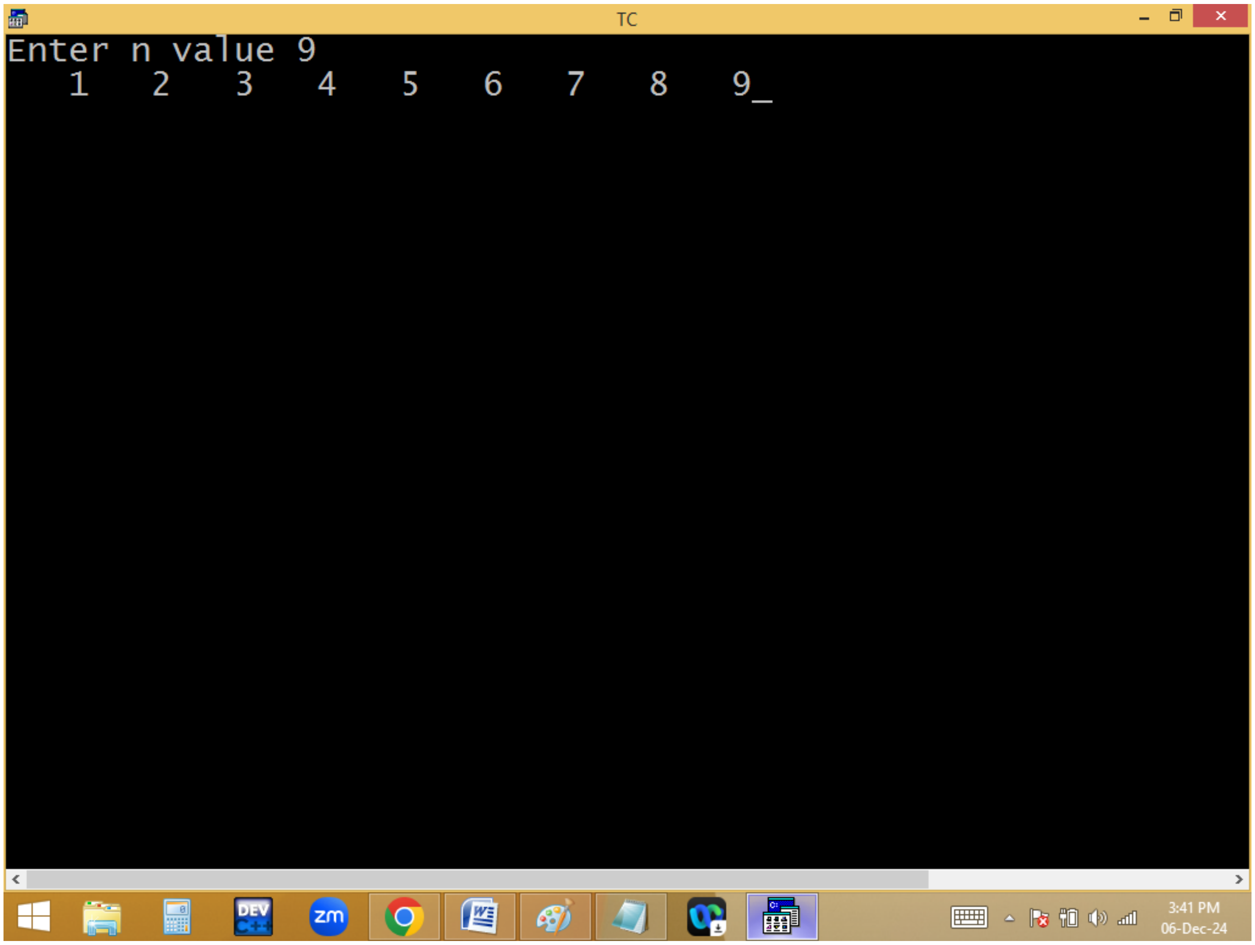

TC
Enter n value 100
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

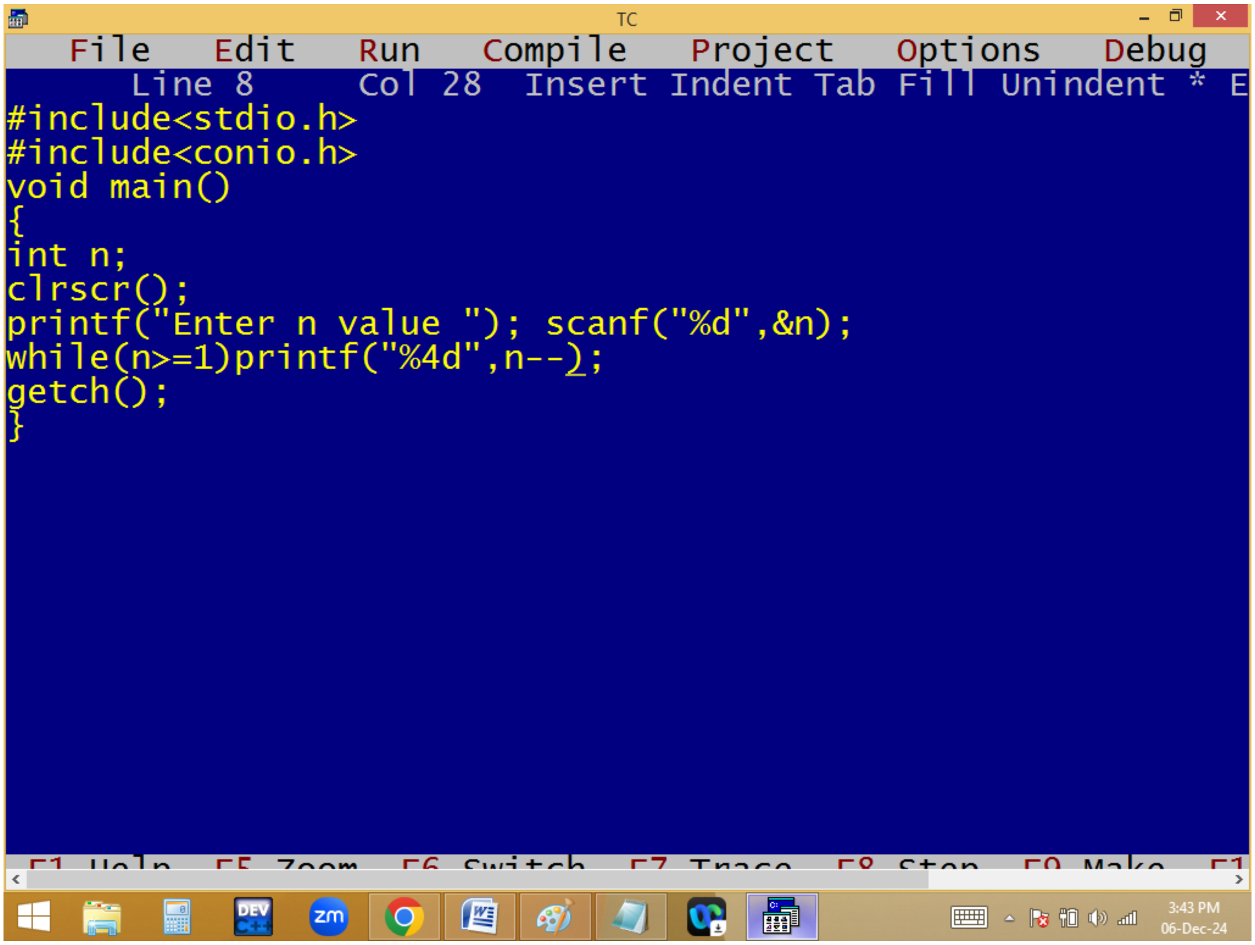
i	<=	n
1		5
2		
3		
4		
5		
6		

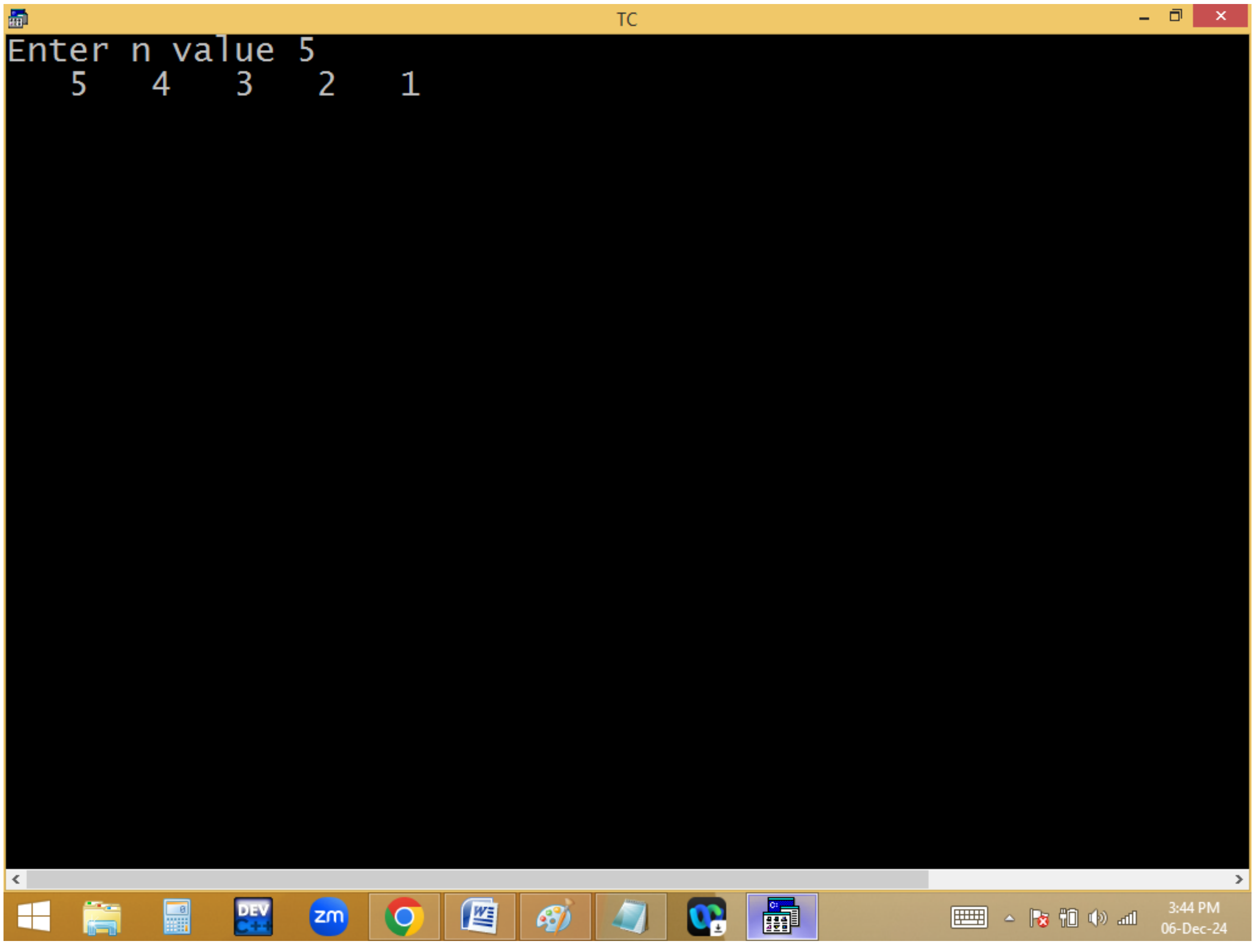
```
while( i<=n )
{
p("%4d", i );
i++;
}
```




TC
File Edit Run Compile Project Options Debug
Line 8 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1, n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(i<=n)printf("%4d",i++);
getch();
}
3:41 PM
06-Dec-24


TC
Enter n value 9
1 2 3 4 5 6 7 8 9_
3:41 PM
06-Dec-24


TC
File Edit Run Compile Project Options Debug
Line 8 Col 28 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value "); scanf("%d",&n);
while(n>=1)printf("%4d",n--);
getch();
}
3:43 PM
06-Dec-24


TC
Enter n value 5
5   4   3   2   1
3:44 PM
06-Dec-24